Lab 7: Linux Capabilities and Least Privilege

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Lab 6 showed privilege as all-or-nothing: a process is root or it is not, and setuid grants the whole of root when perhaps a sliver was needed. Linux **capabilities** decompose root into many independent rights, so a process can be granted exactly the one privilege it requires and nothing more. This lab has you enumerate the capabilities active on a running system and inside containers, attach a **file capability** to a binary instead of making it setuid root (the fix you reached for in Lab 6), and apply least privilege to a real service. The point throughout: **splitting root is the right idea, and it is harder to do correctly than it looks.**

Safety. No self-lockout risk here, but you will attach and remove file capabilities on binaries. Snapshot `lab7-clean` before you start. Pull the container images in the pre-lab. Exercises 3 and 5 also run `apk add`, so the session needs network access.

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Learning Objectives

By completing this lab, students will be able to:

1. Name the capability sets (permitted, effective, inheritable, ambient, bounding) and read a process's `CapEff` from `/proc`.
2. Enumerate the capabilities a process or container holds and decode the bitmask.
3. Attach a **file capability** to a binary and explain why it is a smaller grant than setuid root.
4. Explain why `--privileged` under rootful Docker is effectively host root, and why "runs as non-root" is not "is sandboxed" (a non-root UID alone does not prevent acquiring a capability). *(The full "capabilities alone are not containment" argument needs namespaces and seccomp and is completed in Lab 12; here you only reach the "non-root is not sandboxed" observation of Exercise 3b.)*
5. Apply least privilege to a service by granting the single capability it needs.

Capability enumeration and dropping are woven into the capstone's container-reaching link.

What from this lab is assessed. Two things, both handed in with **Lab report 3** at the end of the block (Lab 8): your answers to the preparation questions **P1 and P2**, and your **Reflection**. The exercises themselves are not handed in. Keep your capability transcripts: Lab 11 and the capstone reuse the least-privilege reasoning you build here.

Before the lab (at home, ~30-45 min)

Boot the VM, snapshot `lab7-clean`, re-read Lecture 7 if you have already had it. Pull the images the container exercises use so the session is not spent waiting:

```
docker pull alpine
docker pull python:3.12-alpine
```

Unzip `lab7-files.zip` into `~/ssc-lab7` and run every command of this lab from there (it contains `lab7-bind80.c`, the bind helper you compile in Part B; `captest.py`, a small bind-a-socket probe; and `Dockerfile.capbind`, the image you build in Exercise 5). Confirm `capsh` and `getcap`/`setcap` are present:

```
which capsh getcap setcap        # from libcap; installed on the image
```

Preparation questions (submit with Lab report 3)

- **P1.** Linux splits the power of root into separate capabilities. About how many are there? Name two of them and say what each one allows.
- **P2.** A program can get privilege from a file capability or from the setuid-root bit. In one sentence, why is the file capability the smaller grant?

Part A: Capabilities on the running host

Exercise 1. Read the capability sets of your own shell, then of a privileged process. These commands are probably new to you, so first what you are looking at:

- **`/proc/self/status`** is the kernel's report on the process that reads it (here `grep`, whose sets match your shell's in this exercise). `grep Cap` keeps five lines, one per capability set.
- **Each value is a bitmask in hexadecimal**, one bit per capability. `0000000000000000` is an empty set; `000001ffffffffff` has the lowest 41 bits on, which is every capability on this kernel.
- **`capsh --decode=<mask>`** turns a mask into capability names.

Start with your own shell, decoding two of its sets:

```
grep Cap /proc/self/status
capsh --decode=$(grep CapEff /proc/self/status | awk '{print $2}')
capsh --decode=$(grep CapBnd /proc/self/status | awk '{print $2}')
```

Now a system daemon running as root:

```
pid=$(pgrep -n cron || pgrep -n systemd)
sudo grep Cap /proc/$pid/status
capsh --decode=$(sudo grep CapEff /proc/$pid/status | awk '{print $2}')
```

1a. Answer these questions:

- Compare the `CapEff` of your shell with the `CapEff` of the daemon. What is different?
- Name the five capability sets.
- In one sentence, what does `CapEff` (effective) control?
- In one sentence, what does `CapBnd` (bounding) control?
- When a process makes a syscall, the kernel checks the *effective* set, not the *permitted* set. Why?

Part B: File capabilities instead of setuid

In Lab 6 you tried, among other options, replacing the setuid `backup-tool`'s blanket root with the specific capability it needs, and measured that it narrowed the grant without closing the hole. Here you apply the same mechanism to a program whose one need it does fit, and see the difference.

Exercise 2. Take a program that needs one privilege and grant it that privilege two ways: setuid root, then a file capability. Compare.

`lab7-bind80.c`, from `lab7-files.zip`, tries to bind TCP port 80 and prints the result with its `uid` and `euid`. On this VM ports below 1024 are privileged, so an ordinary process should be refused. Compile it, then run it:

```
gcc -o ~/capbind lab7-bind80.c
~/capbind
```

The compile prints nothing; the message comes from running the program.

The blunt way: setuid root (delete this copy when you finish the exercise). Make a copy, give it to root and set the setuid bit:

```
sudo cp ~/capbind ~/capbind_suid
sudo chown root ~/capbind_suid
sudo chmod u+s ~/capbind_suid
```

Try to run the program again:

```
~/capbind_suid
```

The sharp way: one file capability, no setuid. `cap_net_bind_service` is the capability that allows binding a privileged port. In `+ep`, the `p` asks for it in the file's permitted set and the `e` turns on the file's effective bit, a single flag that makes the capability active as soon as the program starts. Exercise 6 comes back to that flag.

```
sudo cp ~/capbind ~/capbind_cap
sudo setcap cap_net_bind_service+ep ~/capbind_cap
getcap ~/capbind_cap
```

Run this copy:

```
~/capbind_cap
```

Read `uid` and `euid` in all three runs. The contrast the exercise wants is between three runs of one program. Setting or removing a file capability needs `sudo`.

2a. Both copies can bind port 80, but they give the program very different power.

- In the setuid run, what is the effective UID of the process?
- What can a process with that UID do on this VM?
- In the file-capability run, does the process have any privilege beyond binding a privileged port?
- What does the setuid copy get that the file-capability copy does not get? (Giving a program only what it needs is called ”least privilege”.)

2b. Remove the capability from `~/capbind_cap`. The command is not given here: find it in `setcap(8)`.

- Which command did you use?
- Run `~/capbind_cap` again. Does the bind now fail?
- Why is this command the right way to take the capability away, rather than deleting the file or changing its mode bits with `chmod`?

The `ping` program faced the same choice as `capbind`. Traditionally, `ping` opens a raw socket, which is a privileged operation, so for years it was installed setuid root.

- Run `getcap /usr/bin/ping` on the course VM. What does it show?
- Why does `ping` on the course VM not need the setuid-root bit?

Part C: Capabilities inside containers

The arc of Parts C, D and E. One question in three steps: *how much of root is a container's root* (Part C), *how small can a service's grant be* (Part D), and *how do you run that service as non-root with exactly*

the one capability it needs (Part E). Each step corrects the one before, so keep the destination in view: Exercise 3 reads the default and privileged sets; 3b adds a non-root UID; Exercise 4 looks for the minimal grant; Exercise 5 makes it genuinely non-root; and Exercise 6 matches the ceiling (`CapBnd`) to the grant and shows what breaks when it drops too far.

Exercise 3. Compare the default and privileged container capability sets.

A process in a container is an ordinary process on the host's kernel, and capabilities apply to it as they do to any other. So the question that matters is what a container's root can do to the host. Docker starts containers with a reduced set; the `--privileged` flag asks for no reduction. You read the five sets in each case and decode `CapEff` to see how much of root a default container keeps and what `--privileged` changes.

The default container:

```
docker run --rm alpine sh -c "grep Cap /proc/self/status"
docker run --rm alpine sh -c \
  "apk add -q libcap && capsh --decode=\$(grep CapEff /proc/self/status | awk
   ↪  '{print \$2}')"
```

The privileged container:

```
docker run --rm --privileged alpine sh -c "grep Cap /proc/self/status"
docker run --rm --privileged alpine sh -c \
  "apk add -q libcap && capsh --decode=\$(grep CapEff /proc/self/status | awk
   ↪  '{print \$2}')"
```

3a. Docker's default set has 14 capabilities. `--privileged` gives every capability the host kernel has. On the course VM, the Docker daemon runs as root (rootful Docker).

- Record the output of all four commands. The privileged mask is a value you already saw in Exercise 1.
- Explain why a `--privileged` container on the course VM is, in practice, **root on the host**. Capabilities are part of the answer, but `--privileged` also opens the host's devices and removes other restrictions.
- Name two capabilities that `--privileged` adds to the default set, and say what each one allows.
- An operator may run `--privileged` and think it is safe. What wrong belief could lead to that?

3b. Is running as a non-root user enough to make a container a sandbox? Measure before you answer: run the first command of Exercise 3 again (the default container's `grep Cap`), with `--user 1000` added before `alpine`.

One definition first. `CapBnd`, the bounding set, is what this lab calls the ceiling. When a program starts with `execve`, the kernel applies it as a mask to the file's permitted capabilities. It does not take away capabilities that the process already holds.

- Record `CapPrm`, `CapEff` and `CapBnd`.
- What does the process hold, compared with the run as root in the default container?
- What is left in `CapBnd`?

What the process holds now is only part of the answer. Think back to Part B:

- Name one way a process in this container could still get one of the capabilities left in `CapBnd`.
- What must exist in the image for that to work?
- Name one capability left in `CapBnd`, and say what it would allow.
- What does this mean for treating ”runs as non-root” as if it meant ”is sandboxed”?

Part D: Least privilege for a real service

Exercise 4. Find the smallest grant that lets a service bind a privileged port, and prove that anything less fails.

The service is `captest.py`, from `lab7-files.zip`: it tries to bind the port you give it and prints the result with its `uid`. Read the file before you run it. You run it in containers with Docker's `--cap-add` and `--cap-drop`. In every test, read the UID as well as the result, because the goal is a **non-root** service.

The kernel setting `net.ipv4.ip_unprivileged_port_start` is the lowest port an unprivileged process may bind. Its classical value is 1024, the privileged-port boundary, and below it the capability is what decides. Docker changes this value inside containers (4b asks how and why), so every test passes a flag that puts the classical value back. Keep the flag in a variable. It lives only in this shell: in a new terminal, set it again, or `$S` is empty and a bind can succeed for the wrong reason.

```
S="--sysctl net.ipv4.ip_unprivileged_port_start=1024"
```

Test 1: root, with Docker's default capabilities.

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S python:3.12-alpine python
 ↪  /captest.py 80
```

Test 2: non-root, with no extra grant.

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --user 1000 python:3.12-alpine
 ↪  python /captest.py 80
```

Test 3: drop every capability and add back exactly `NET_BIND_SERVICE`.

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --cap-drop ALL --cap-add
 ↪  NET_BIND_SERVICE python:3.12-alpine python /captest.py 80
```

Test 4: the negative control the word ”minimal” needs. Drop every capability and add nothing back.

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --cap-drop ALL
 ↪  python:3.12-alpine python /captest.py 80
```

Tests 3 and 4 differ by one capability. That pair is what turns ”minimal” from a claim into a demonstration: without test 4 you have shown that a configuration works, not that the smaller one does not.

The result is not the whole story, so look at what the flags did to the sets. First with no flags, then with the flags of test 3:

```
docker run --rm python:3.12-alpine python -c
 ↪  "print(open('/proc/self/status').read())" | grep Cap
docker run --rm --cap-drop ALL --cap-add NET_BIND_SERVICE python:3.12-alpine python
 ↪  -c "print(open('/proc/self/status').read())" | grep Cap
```

4a. Record the result and the UID of tests 1 to 4. Then answer:

- Of the four tests, which one binds with the smallest grant? Give its flags: which capabilities it drops and which one it adds.
- What did those flags do to `CapBnd`?
- Does that test meet the non-root goal?
- Compare `--cap-drop ALL --cap-add NET_BIND_SERVICE` with Docker's default set. Is it a smaller or a larger grant?
- Why is it safer to start from nothing and add, than to start from the default set and remove?

4b. Your tests set `ip_unprivileged_port_start` to 1024; Docker's own default inside a container is `0`. Run test 2 again without the flag:

```
docker run --rm -v "$PWD/captest.py:/captest.py" --user 1000 python:3.12-alpine
 ↪  python /captest.py 80
```

This command has no `$S` on purpose.

- What does the value `0` allow a process to do?
- Why do you think Docker uses this value by default?
- In a container started without the flag, does `CAP_NET_BIND_SERVICE` decide whether a process can bind port 80?

- What does this tell you about reasoning only from capabilities, without checking the rest of the configuration?

Part E: A non-root service with one capability

Look again at the UID in test 3. It binds as `uid=0`: root inside the container, holding one capability instead of the default fourteen. That is a real reduction, and it is not what Exercise 4 set out to find: it asked for a *non-root* service, and none of the four tests gives one that binds.

Exercise 5: make it actually non-root. The answer is not a Docker flag, it is the mechanism you already used in Part B, moved inside the image. `Dockerfile.capbind`, in `lab7-files.zip`, is `python:3.12-alpine` with one line added: `setcap cap_net_bind_service+ep` on the interpreter. Build it:

```
docker build -t lab7-capbind -f Dockerfile.capbind .
```

Run the service from your image as an ordinary user, with no `--cap-add`, then ask the interpreter what it is holding:

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --user 1000 lab7-capbind python
 ↪  /captest.py 80
docker run --rm --user 1000 lab7-capbind python -c
 ↪  "print(open('/proc/self/status').read())" | grep Cap
```

Now the control, which uses a different image: the same two commands on the unmodified `python:3.12-alpine`, which has no file capability.

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --user 1000 python:3.12-alpine
 ↪  python /captest.py 80
docker run --rm --user 1000 python:3.12-alpine python -c
 ↪  "print(open('/proc/self/status').read())" | grep Cap
```

5a. Record the UID and the capability sets for both images. The four sets you need are `CapPrm`, `CapEff`, `CapAmb` and `CapBnd`. `CapInh` is printed too, and it is zero in every run.

- What is the value of `CapPrm` in your image, `lab7-capbind`?
- Which capability does that bit stand for?
- Did the capability reach the process through the ambient set (`CapAmb`)?
- What do you see in `CapBnd` in each image?
- Does the grant apply only to `captest.py`, or to any other Python code this interpreter runs?

Exercise 6: the grant is minimal, the ceiling is not. `CapBnd` in Exercise 5 still carries Docker's default set. In Exercise 4 you saw which flags bring it down, so use them here: non-root, on your image, with the ceiling reduced to the one capability. Bind, then read the sets:

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --user 1000 --cap-drop ALL
 ↪  --cap-add NET_BIND_SERVICE lab7-capbind python /captest.py 80
docker run --rm --user 1000 --cap-drop ALL --cap-add NET_BIND_SERVICE lab7-capbind
 ↪  python -c "print(open('/proc/self/status').read())" | grep Cap
```

Now take one step further and drop the ceiling below what the file capability needs:

```
docker run --rm -v "$PWD/captest.py:/captest.py" $S --user 1000 --cap-drop ALL
 ↪  lab7-capbind python /captest.py 80
```

6a. Record the UID and the capability sets of the service (the first two commands). Then look at the output of the last command.

- What did you expect to happen?
- What happened?
- Explain the result. You need two facts about the file capability, and a section of `capabilities(7)` for each:

  - **+p** puts the capability in the file's permitted set. Read ”Transformation of capabilities during execve()”.
  - **+e** is the file's effective bit: one flag, not a set of capabilities. Read ”Safety checking for capability-dumb binaries”.
- What would change in the last command if the capability had been attached with +p only?

6b. Answer these questions:

- What is the minimal grant that lets a non-root service bind a privileged port?
- On the course VM, why can no combination of `--cap-add` and `--cap-drop` alone produce it?
- How does this connect to what you did in Part B?

Conclusion

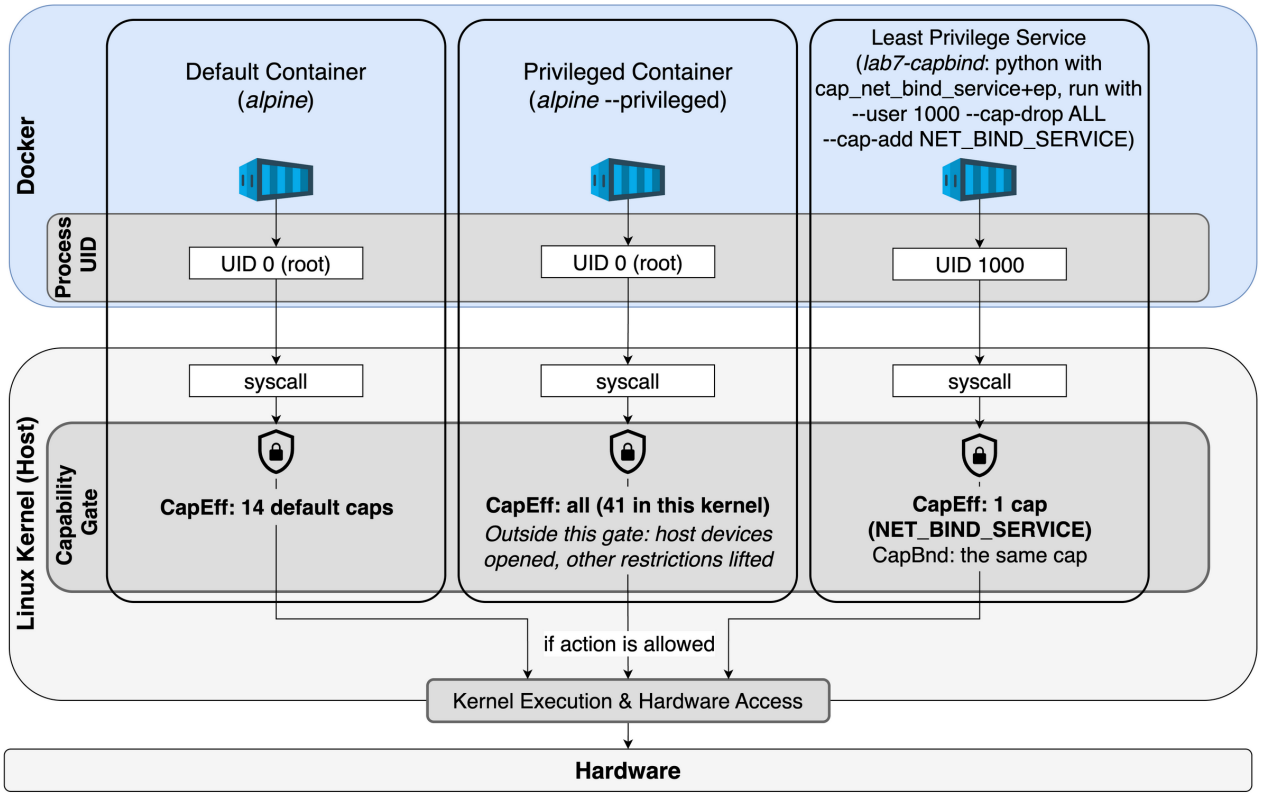
You might expect a non-root process on the unmodified `python:3.12-alpine` image, with `--cap-add NET_BIND_SERVICE` and no file capability, to bind port 80 under `$S` as well. It does **not**. These are the sets measured on the course image:

run on `python:3.12-alpine`	`CapPrm`	`CapEff`	`CapBnd`
`--cap-drop ALL --cap-add NET_BIND_SERVICE`, root inside	`0x400`	`0x400`	`0x400`
the same, plus `--user 1000`	`0`	`0`	`0x400`

These are two end states, not a sequence: they do not show the order in which the runtime built the sets, and another runtime may build them differently. In the second run the capability is in the bounding set but not in permitted or effective. Only the effective set is checked at the bind, so under `$S` the bind is refused. Without `--cap-drop ALL`, `--cap-add` changes nothing at all: `NET_BIND_SERVICE` is already in Docker's default bounding set, and `CapPrm` and `CapEff` stay at zero.

So with flags alone, the smallest grant is one capability held by root inside the container. A non-root service on this image needs a file capability on the binary (Parts B and E) or Docker's default open ports (4b).

The figure puts three of the cases you measured side by side: the default and privileged containers of Exercise 3, and the service of Exercise 6. In that third case the one capability comes from the file capability on the interpreter; the `--cap-drop` and `--cap-add` flags only lower the ceiling to match it. In all three, when the process asks for a privileged operation, the kernel checks the `CapEff` it holds at the same gate. `--privileged` also changes things outside that gate: it opens host devices and lifts other restrictions (3a).



Copy your capability transcripts off the VM, then restore `lab7-clean`.

If it does not work

Both of these look like the capability did nothing. Neither is a broken lab.

What you see	What to do
`--user 1000 --cap-add NET_BIND_SERVICE` still cannot bind port 80 under `$S`	Measure it rather than assume: that run ends with `CapPrm = CapEff = 0` and `CapBnd` at Docker's default set, so the capability is in the bounding set and nowhere the process can use. `--cap-add` is a no-op here: `NET_BIND_SERVICE` is already in that default set. Use `--cap-drop ALL --cap-add NET_BIND_SERVICE` as root inside, or give the binary a **file capability** (Exercise 5).
The privileged-port boundary seems not to exist in a container	It does not, by default: `net.ipv4.ip_unprivileged_port_start` is 0 inside. Pass `--sysctl net.ipv4.ip_unprivileged_port_start=1024` to restore the boundary you are trying to observe.

If none of these matches, read the error message in full first: it usually names the layer that refused, and that is most of the answer. Then ask your lab teacher, in the session or by email.

Reflection

Your reflection goes into Lab report 3, as an appendix, beside the reflections from the other labs in this block. They are assessed with the report: every one has to be there, each answering its questions with a position you argue from what you did in the lab.

In a short piece (200-300 words in all, about 100 for each question):

1. Capabilities exist so that least privilege is possible. Lab 6's setuid binary and this lab's file-capability copy meet the same need, but if either one is exploited, the damage is very different. So why is setuid still everywhere? What makes capabilities harder to adopt in practice than the idea suggests?
2. You saw two ways around a capability restriction: `--privileged` gives every capability back, and Docker's port setting removes the check altogether. In general, why is "I dropped the dangerous capability" not the same as "the dangerous action is prevented"?
3. Capstone link. In the capstone, one step of the attack works because a container keeps a capability it should not have. You have not seen that container yet. From the capabilities you named in Exercise 3a, which one would you look for first and remove, and why? Which capability sets of a process in the container would you read to check that it was removed?

References

- `capabilities(7)` (the full list and the five sets); `capsh(1)`, `getcap(8)`, `setcap(8)`.
- Docker default capability set: the `--cap-add`/`--cap-drop`/`--privileged` documentation.
- `ip_unprivileged_port_start` and the reserved-port change in Docker 20.10.
- W. Du, *Computer Security: A Hands-on Approach*, 3rd ed., 2022, the capabilities material.

Deliverables

There is **no separate report for this lab**; keep your capability transcripts. Block 3's assessed deliverable is **Lab report 3**, produced in Lab 8, which collects this lab's preparation questions P1-P2 and its Reflection. The least-privilege reasoning here (4a, and the "start from nothing and add" principle) is directly reused in Lab 11's compose-hardening report and in the capstone.